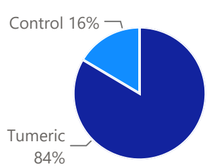
# Cancer Survival ML Project | Cohort Presentation

- 1,208 patients with tumor data were filtered to 389 with complete survival, stage, and age records. Higher cancer stages indicate more advanced tumors
- 5-year survival was determined only from confirmed follow-up or time to death. Patients with unknown or insufficient follow-up were excluded.

## PATIENT COUNT

389

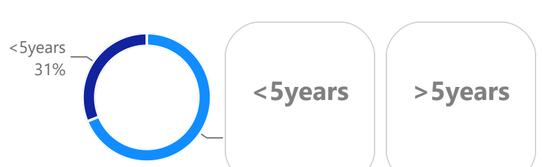
## TUMOR/CONTROL RATIO



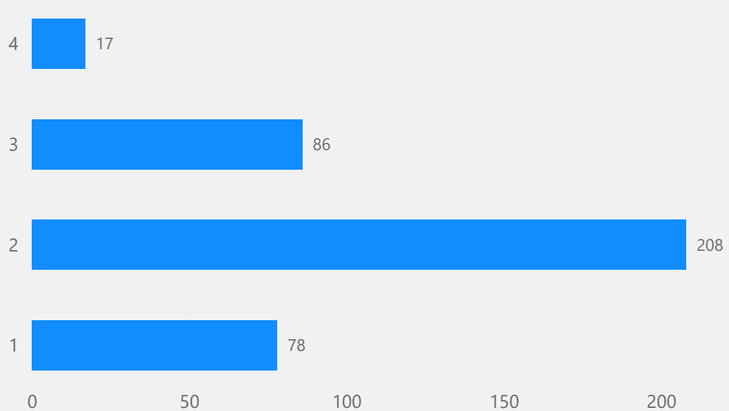

## MEDIAN PATIENT AGE

57.86

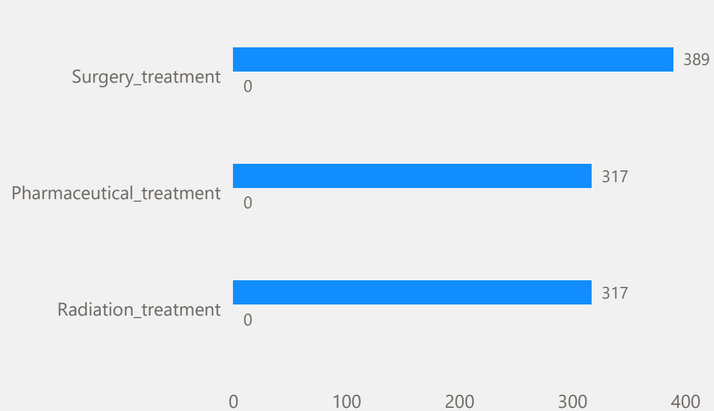
## PATIENT 5 YEARS SURVIVAL RATIO



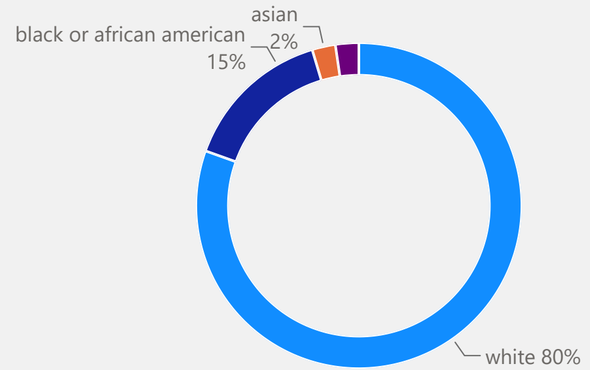

## Patients distribution per cancer stages




## Patients distribution per treatment type




## Patients origin distribution



# Cancer Survival ML Project | Pipeline Overview

- End-to-end pipeline: public TCGA-BRCA data → cloud storage → SQL parsing → statistical gene selection → neural network → dashboard.
- 1,208 patients narrowed to 389 with complete survival, stage and age records (325 tumor + 64 control samples).
- Each step is a numbered notebook in the GitHub repository

## Key data decisions:

- Metastatic samples **excluded in the API query**.
- **11 patients** with several tumor samples **reduced to one**, so no patient appears in **both train and test**.
- GDC moved follow-up data to a new field, so I queried the new one.
- Patients with **unknown 5-year outcome** were **excluded**, not assumed to survive.
- Scaling and PCA were **fit on training data only**.

## DATA MIGRATION

### GDC API Call

Pull case + file metadata; exclude Metastatic samples at source

Python + requests

Call for metadata and RNA-seq data **(1,208 raw files gathered)**

### S3 Upload

Stream raw RNA-seq count files to AWS S3

boto3 + AWS S3

Upload of raw RNA-seq files, **both Tumor and Control cohorts**

### Metadata Parsing

DuckDB: unnest JSON, compute survival / stage / age

DuckDB SQL

**389 profiles selected** based on metadata's completeness

### Parquet Conversion

Filter to protein-coding genes per sample

DuckDB + Parquet

Count files from s3 selected for protein-coding genes, and merged samples into one Parquet file/group.
**Metric: 389 samples × 19,962 genes**

## DATA ANALYSIS & MODELING

### DESeq2

Differential expression, tumor vs. control

PyDESeq2

**Selection of 1,737 significant genes.** (adjusted $p < 0.05$, $|\log2FC| > 2$)

### Feature Engineering

PCA (50 components) vs. DESeq2-selected genes

scikit-learn PCA

The expression input: **7 clinical metadata features** with either **50 PCA components or DESeq2-selected genes.**

### Neural Network

Dual-branch model, 5-fold cross-validation

PyTorch

The two branches (metadata | RNA-seq) **concatenated (96 units),** passed through **batch norm, ReLU** and **dropout,** then a **single output.**

### Baseline Models

Random Forest & Logistic Regression comparison

scikit-learn

**Logistic Regression | Random Forest:** Randomized search **over 20 draws** with **5 fold CV,** scored on **accuracy**

### Power BI Dashboard

Cohort profile + model performance

Power BI

**Final deliverable**

# Cancer Survival ML Project | Model Performance

- Goal: predict from clinical data and gene expression whether a patient survives at least 5 years after diagnosis. The model is trained on the 325 tumor samples of the cohort.
- Two ways of condensing ~19,000 genes into a usable input are compared: PCA and DESeq2 (see buttons).
- Top row: Neural network, 5-fold cross-validation: mean ± standard deviation across folds. Lower sections: one train/test split, used to show how the model behaves.

| AUROC | MACRO F1 | ACCURACY |
|---|---|---|
| 0.80 ±0.067 | 0.69 ±0.046 | 0.72 ±0.039 |
| vs PCA ▼ 0.044 | vs PCA ▼ 0.058 | vs PCA ▼ 0.058 |
| Ranking quality across all cutoffs | Balanced score across both groups | Share of correct predictions |

Deseq2 | PCA

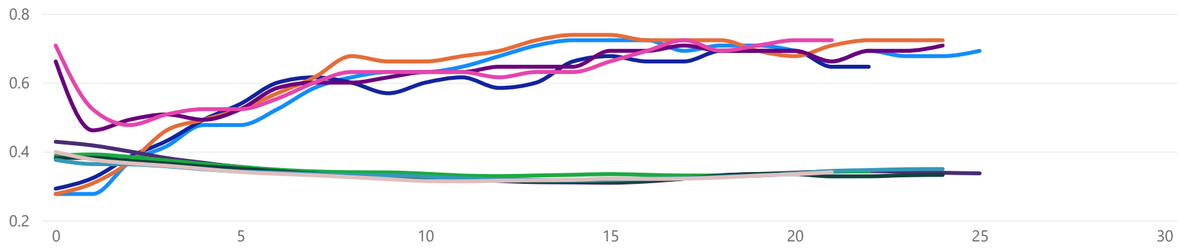
Bookmark toggle - Select the gene-input method (updates the whole page)

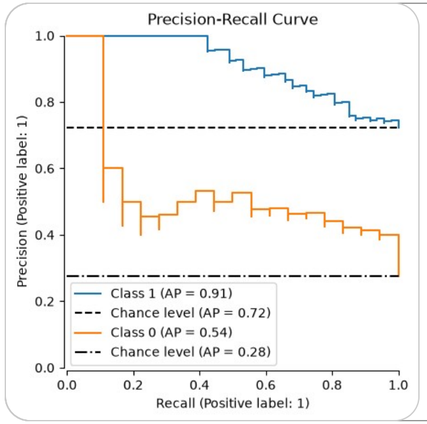
**PCA**: compresses all genes into 50 summary components that capture most of the variation between patients. It never looks at survival.
**DESeq2**: keeps only the genes that differ significantly between tumor and normal tissue (adjusted p < 0.05, |log2 fold change| > 2). Chosen by biology, not by survival.

## Model behavior - Single Train/Test split

Learning curves: accuracy (upper lines) and loss (lower lines) by training epoch

Each line is one of 5 training runs. Accuracy rises and levels off while loss falls and flattens; training stops early if the model stop improving.




- **Precision** defines how often a patient flagged by the model truly belongs to the positive group.
- **Recall** defines the model capacity to flag the right patients across the whole cohort.
- **AP (average precision)** summarizes the curve, while dashed lines show random guessing. The orange line (Class 0 - non-survivors), are the hardest group to define, due to low number and noisy estimates.

| Recall Class 0 | Recall Class 1 |
|---|---|
| 0.69 ±0.100 | 0.73 ±0.034 |
| vs PCA ▼ 0.035 | vs PCA ▼ 0.044 |

## Model comparison - Matching patients cohort (65 samples)

**Neural network confusion matrix**

| Actual | < 5 yrs | > 5+ yrs |
|---|---|---|
| < 5 yrs | 13 | 5 |
| > 5+ yrs | 15 | 32 |

**Logistic Regression confusion matrix**

| Actual | < 5 yrs | > 5+ yrs |
|---|---|---|
| < 5 yrs | 7 | 11 |
| > 5+ yrs | 6 | 41 |

**Random Forest confusion matrix**

| Actual | < 5 yrs | > 5+ yrs |
|---|---|---|
| < 5 yrs | 1 | 17 |
| > 5+ yrs | 1 | 46 |

- **Neural Network:** finds 13 of 18 non-survivor (misses 5); 15 of 47 survivors wrongly flagged; accuracy 69%
- **Logistic Regression:** finds 3 of 18 non-survivors (misses 15); 7 survivors wrongly flagged; predicts "survived" for 55 of 65 patients; accuracy 66%
- **Random Forest:** finds 1 of 18 non-survivors (misses 17); 1 survivor wrongly flagged; predicts "survived" for 63 of 65; accuracy 72%

## Overall takeaway:

The neural network ranks patients well (**AUROC 0.85 ± 0.03 with PCA, 0.80 ± 0.07** with DESeq2) and flags **73%** and **69%** of those who died within 5 years. **PCA edges DESeq2**, but only modestly. **Simpler models catch only 1–7 of 18** at-risk patients vs **13–16 for the neural network**, illustrating the need for a model of higher complexity when working with highly dimensional RNA-seq/Metadata.